

The Living Enterprise

Escape Velocity 1.0 · Problem P-03: Multi-Agent Systems that plan, delegate and recover

What P-03 asks for

Most multi-agent demos work on the happy path and collapse the moment a step fails. P-03 asks for a real system: one that breaks a task into steps on its own, delegates to specialised agents, calls real tools, recovers when something goes wrong, knows when to stop, asks a human before risky actions, and leaves a trace a person can audit. In the brief's words, the interesting engineering is in **the recovery, the stopping conditions and the audit trail**, not the number of agents.

The problem we are solving

Every company handles requests like *"Our vendor wants to renew the contract at a higher price."* Answering it well means checking places different teams own: the contract (legal), the procurement policy (procurement), payment and performance records (finance), and live data such as today's exchange rate.

Done by hand, it goes wrong in predictable ways:

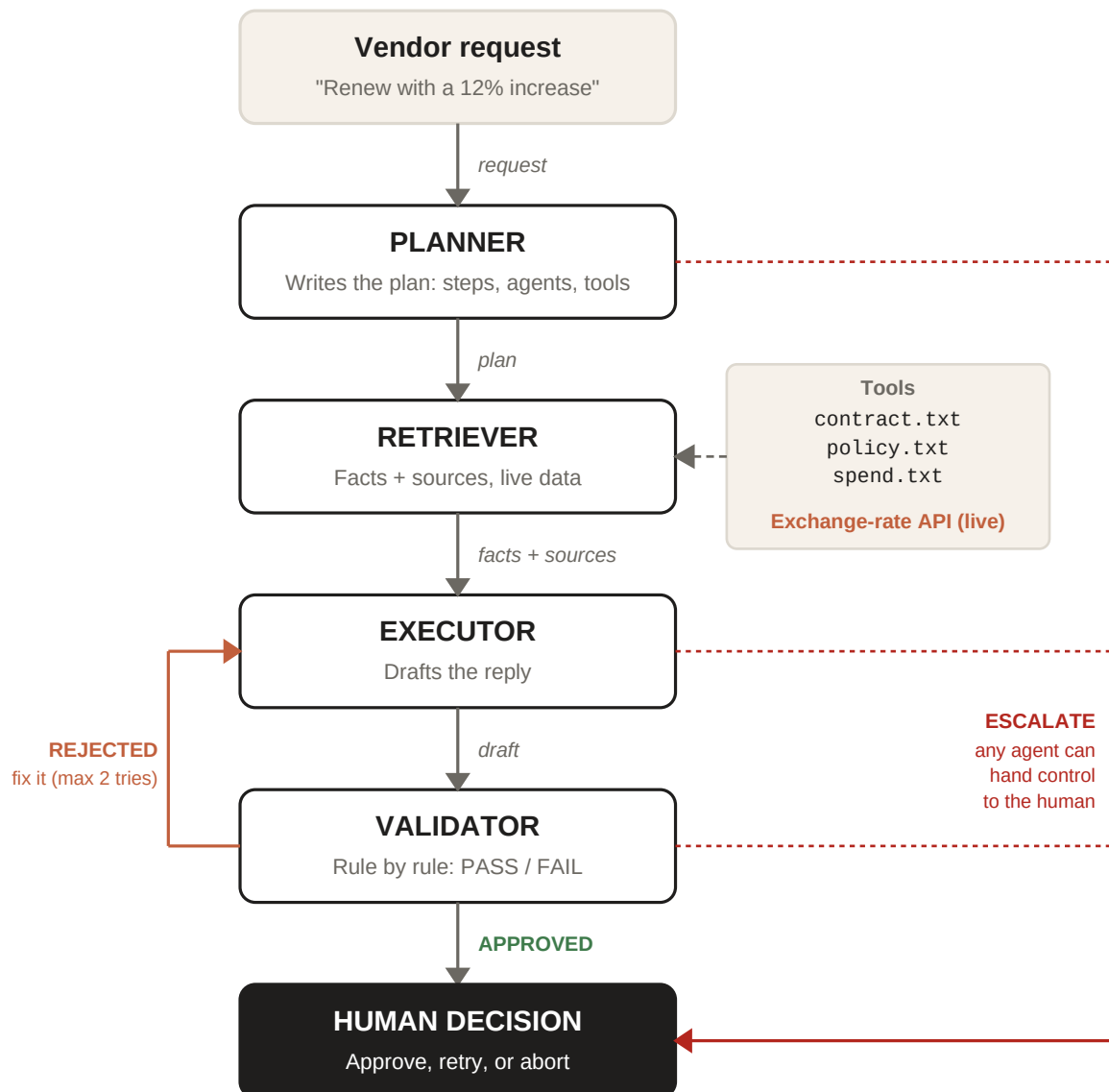
- **It is slow.** The request passes between people and teams, and each handoff adds days.
- **Details get missed.** A price cap buried in Clause 8.2, or refunds the vendor still owes, are easy to overlook.
- **Missed details cost money.** Accepting an increase above the cap, or renewing without claiming credits, means overpaying.
- **A single chatbot is not a fix.** It cannot show sources, nobody checks its work, and it breaks silently when a tool fails.

Today (manual or one chatbot)	With The Living Enterprise
Request bounces between teams	Agents hand work to each other in minutes
Facts looked up from memory	Every fact quoted from a document, with its clause number
No one double-checks the reply	A separate Validator checks every rule and can reject the draft
Breaks silently when a system is down	Retries, falls back, and flags the problem
Fully manual or fully automatic	Automatic, but hands control to a human when it matters
No record of how the decision was made	A full trace of every step, with time and cost

Our idea in one sentence

A small AI office team handles a business request end to end: it adapts its plan to the request, survives failures, knows when to stop, asks a human before acting, and shows its work.

The workflow



The team, and why each agent exists

Agent	Job	Why it is a separate agent	Model
Planner	Reads the request and writes the plan: which steps, which agent, which tool	Different requests need different plans; this is real delegation, not a fixed chain	Sonnet 5
Retriever	Finds facts in company documents and calls the live exchange-rate API	Only agent allowed to touch data, so every fact has a source	Haiku 4.5
Executor	Drafts the reply using only the retrieved facts	Writing is kept apart from fact-finding and checking	Haiku 4.5
Validator	Checks the draft rule by rule and can reject it	A separate critic catches mistakes the writer cannot see in its own work	Sonnet 5
Human	Approves, retries, or aborts	Irreversible actions and unresolved problems need a person	—

The flag system: when control goes to a human

The system fixes what it can on its own and hands control to a human when it cannot. Problems are raised at two levels, and every flag and every human decision is saved in the trace.

Level 1: Warning. The system recovers by itself and tells the human at final approval.

Problem	What the system does
Exchange-rate API slow or fails once	Retries and succeeds
API keeps failing	Uses the saved backup rate, marked "unverified"
Validator rejects the first draft	Executor fixes it; the second draft passes

Level 2: Escalation. The system pauses immediately and the human decides.

Trigger	Why a human must decide
Validator rejects the draft twice	The agents cannot fix it on their own
A required document is missing	Guessing would be dangerous
Documents contradict each other	Needs human judgment
Step, time, or cost limit reached	Decide whether to spend more
Amount above a threshold (e.g. Rs 25 lakh)	High-stakes decision
An agent returns nonsense after retries	Its output cannot be trusted
Any irreversible action (sending the email)	Always needs approval, even with no problems

At an escalation the human sees the reason, the stage, and the problem, then chooses: **retry with instructions**, **accept anyway** (and take responsibility), or **abort**.

How we meet each P-03 requirement

P-03 requirement	How The Living Enterprise meets it
Task needs more than one agent; justify each	Four agents with separate jobs (see the team table)
Real planning and delegation, not a hardcoded chain	Planner writes a different plan for each request type
At least one real external tool or API	Live exchange-rate API, with its real responses checked
Survive failed, slow or malformed steps	Timeout, retry, backup rate, and a chaos mode that breaks the API on purpose
Readable trace	Trace file: which agent, what input, what output, and why
Stopping condition	Limits on steps, rewrites, time, and cost
Human approval for irreversible actions	Nothing is sent until a human approves; escalation flags

Advanced directions we cover: a critic that can reject work; shared state all agents read and write; choosing not to call a tool (e.g. no exchange rate needed for an INR quote); cost and time measured per run; completion rate across 5 repeated runs.

What we use and why

- **CrewAI** defines the agents and runs the flow between them, so we focus on recovery and oversight instead of plumbing.
- **Anthropic API (Claude)** is the brain inside each agent: Sonnet 5 for judgment, Haiku 4.5 for routine work, to keep costs low.
- **A live exchange-rate API** is the real external tool P-03 requires.
- **Text files in data/** stand in for company systems (contracts, policy, finance records).
- **One main file.** Easy to read, debug, and walk through in the pitch. A web UI, if added, gets its own file.

The test scenario

Acme Cloud Services asks to renew with a **12% price increase**. The data hides traps a good team should catch:

- The contract (Clause 8.2) caps renewal increases at **7%**.
- Company policy (Rule 3) says increases above **10%** must not be accepted.
- Acme missed its uptime target in **May and July**, and the service credits owed have not been claimed.

Build plan

Stage	What we build	P-03 requirement
1 · Now	Four agents working together in a fixed order	Basics
2	Planner decides the steps, agents and tools; flag system added	Planning and delegation
3	Live exchange-rate API with timeout, retry, backup rate, chaos mode	Real tool; recovery
4	Validator rejects and sends work back; step, time and cost limits	Critic; stopping conditions
5	Human approval before sending; trace file with time and cost	Human approval; audit trail
6	Diagram, 5-run reliability test, demo script, pitch	Presentation